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Abstract

Primitive-shape recognition can provide compact geometric information for robotic manipulation and remanufacturing, but collecting large labelled sets of real 3D scans is expensive. This project investigated the synthetic-to-real generalisation of a PointNet++ classifier trained on CAD-generated artificial primitive-shape (APS) point clouds. Three primitive classes were considered: box, cylinder and sphere. Three APS training conditions were evaluated: clean synthetic point clouds (APS-clean), perturbed synthetic point clouds (APS-error), and a mixed set containing both. The trained models were evaluated on YCB-28, a test set of 560 point clouds derived from 28 real everyday objects in the YCB object and model set, and on a 486-sample primitive-like subset of the ModelNet40 CAD dataset. The final model used 1000 XYZ coordinates per sample, hierarchical PointNet++ set-abstraction layers and AdamW optimisation. Across nine independent runs, the mixed XYZ-only model achieved 89.19% mean classification accuracy on YCB-28, compared with 67.18% for the earlier mixed PointNet++ model using XYZ plus surface normals. For the final mixed model, class-wise accuracies on YCB-28 were 88.70% for boxes, 87.36% for cylinders and 90.91% for spheres. Pairwise Mann–Whitney U tests after Holm correction showed significant differences among the clean, error and mixed training conditions. On the ModelNet40 primitive-like subset, the mixed model achieved 81.00% mean accuracy using ten-vote test-time evaluation. As an alternative optimisation method, RProp achieved 71.20% mean accuracy on YCB-28 and did not improve on AdamW. Overall, the mixed-trained XYZ-only AdamW pipeline produced the strongest external transfer, while the ModelNet40 evaluation also highlighted the limitations of mapping semantic CAD categories to only three primitive classes.

Keywords: point clouds, PointNet++, primitive shape recognition, synthetic-to-real generalisation, YCB.

1. Introduction

1.1 Background and Motivation

Three-dimensional point clouds are increasingly used in robotics because they preserve geometric information that can be lost in two-dimensional imagery. For manipulation, disassembly and remanufacturing, a complete semantic identity is not always necessary: an approximate primitive description such as box, cylinder or sphere can already constrain grasp selection, pose reasoning and task planning. Zheng and Castellani investigated this concept using PointNet trained on artificial primitive-shape (APS) point clouds consisting of computer-generated boxes, cylinders and spheres [9]. They evaluated synthetic-to-real transfer using point clouds derived from real everyday objects in the Yale-CMU-Berkeley (YCB) object and model set [9,13]. Their work was motivated by automated remanufacturing, where object appearance and condition can vary and geometric information may transfer more reliably than texture. Related mechanical-part recognition work further supports the relevance of point-cloud learning to robotic disassembly [12].

This project follows the same general synthetic-to-real approach but replaces PointNet with PointNet++, which learns hierarchical local geometry [2]. It compares three APS training conditions—clean synthetic data, perturbed synthetic data, and a mixed set containing both—and evaluates an earlier XYZ-plus-normal representation against the final XYZ-only pipeline. Generalisation is assessed primarily using YCB-28 and additionally using a manually selected primitive-like subset of the ModelNet40 CAD dataset [14]. Repeated-run statistical analysis and a supplementary comparison with the RProp optimisation algorithm are also included. The central question is whether geometry learned from synthetic primitive shapes can transfer consistently to less ideal objects from independent datasets.

1.2 Problem Statement

Point clouds are unordered, irregular and non-grid data. Their density can vary locally, and real objects often deviate from ideal geometric primitives through handles, tapering, rounded edges, cavities or compound structures. A classifier trained only on mathematically clean synthetic shapes may therefore achieve high accuracy on synthetic data while performing less reliably on realistic objects. This synthetic-to-real gap is important when synthetic training is used to reduce the cost of collecting and labelling large real datasets. The central problem in this project was therefore not simply to maximise accuracy on APS data, but to develop a reproducible three-class PointNet++ classifier that retained strong performance when evaluated on independent object datasets with different geometry and sources.

1.3 Aim and Objectives

The aim was to evaluate PointNet++-based primitive-shape recognition from 3D point clouds, with particular emphasis on generalisation from artificial training data to real-object geometry.

The objectives were to: (i) implement a three-class PointNet++ classifier for box, cylinder and sphere recognition; (ii) prepare APS clean, error and mixed training conditions and controlled validation and test sets; (iii) compare XYZ-plus-normal and XYZ-only input pipelines; (iv) test the trained models on YCB-28 without using YCB data for training or model selection; (v) evaluate generalisation on a primitive-like subset of ModelNet40; (vi) compare AdamW with RProp as an alternative optimisation method; and (vii) quantify variability using repeated runs, descriptive statistics and non-parametric significance tests.

1.4 Scope and Limitations

The study is deliberately limited to three primitive classes and does not claim to provide a universal object-recognition system. YCB-28 is the primary real-object evaluation, while ModelNet40 is used only through a manually defined primitive-like subset because the trained classifier has three outputs rather than forty. The ModelNet40 sphere mapping is particularly weak because the dataset contains no true ball or sphere category; bowl was therefore used only as a weak sphere-like proxy. The APS validation and test sets used in the final pipeline are balanced APS-style sets generated separately from the original 591-sample training data rather than an official benchmark split. The RProp comparison was restricted to the mixed APS condition after mixed training had produced the strongest AdamW results. These design choices are reported explicitly so that the results are interpreted as controlled experimental evidence rather than standard benchmark scores.

2. Literature Review and Technical Background

2.1 3D Point Clouds and Primitive Shape Recognition

A point cloud represents a surface as an unordered set of spatial samples, commonly XYZ coordinates and sometimes additional attributes such as colour or surface normals. Unlike pixels or voxels, points have no fixed neighbourhood topology. Permutation invariance and local geometric relationships must therefore be handled by the learning method [1-3].

Primitive recognition reduces complex object geometry to a small vocabulary such as boxes, cylinders and spheres. This abstraction is useful when downstream manipulation depends more on gross shape than semantic identity. Baronti et al. used geometric fitting to recover primitive shapes from point clouds [11], while Zheng and Castellani showed that a learned classifier could recognise primitive-like real objects after training on artificial examples [9]. The present work follows the latter classification formulation and uses the primitive label as a compact geometric descriptor rather than a complete object model.

2.2 Deep Learning for 3D Point Clouds

Graph-based and attention-based methods such as DGCNN [4] and Point Transformer [7] can learn detailed relationships between neighbouring points, but they introduce additional architectural complexity. PointCNN [5] and SO-Net [6] provide other approaches to learning

local point-cloud structure. PointNet++ was selected for this project because it learns local geometric features at multiple scales while remaining closely related to the original PointNet framework used in the reference APS-to-YCB experiments [2,9]. This allowed the previous synthetic-to-real concept to be extended without moving to a substantially different model family.

PointNeXt has shown that PointNet++-based models can achieve stronger performance through improved training and model scaling [8]. However, introducing these additional architectural and training changes would have made it more difficult to determine which parts of the experimental pipeline were responsible for differences in transfer performance.

2.3 PointNet and PointNet++ Architectures

PointNet was the first widely adopted architecture to operate directly on unordered point sets [1]. It applies a shared multilayer perceptron to individual points and aggregates the resulting features with a symmetric function, allowing the network output to be invariant to input ordering. This global aggregation is effective for classification but does not explicitly model fine local neighbourhoods. PointNet++ addresses that limitation by recursively sampling points, grouping local neighbours and applying PointNet-style feature extraction at increasingly larger spatial scales [2].

The hierarchical principle is appropriate for primitive recognition because features such as planar patches, local curvature and elongated cylindrical structure are fundamentally local before they contribute to a global object description. In the implementation used here, two local set-abstraction stages were followed by a global abstraction stage and fully connected classification layers. The final network used only XYZ coordinates. This differs from the original Zheng and Castellani study, which used the original PointNet rather than PointNet++ [9]. The present work should therefore be viewed as an extension of the APS-to-YCB experimental concept, not as an exact replication.

2.4 Synthetic-to-Real Generalisation in Point-Cloud Recognition

Training on synthetic geometry reduces the cost of collecting labelled real scans, but it creates a domain gap. Artificial primitives have regular surfaces and controlled sampling, whereas real objects can be incomplete, rounded, compound, noisy or only approximately primitive.

Zheng and Castellani observed this problem directly. PointNet trained on APS-clean achieved a median classification accuracy of 78.84% on the YCB-28 test set, while training with error-corrupted APS increased the median to 85.98% [9]. This suggests that controlled perturbation can expose a classifier to geometric deviations that better resemble real objects. The present project extends this idea by comparing clean, error and mixed APS training and by investigating whether the final XYZ-only pipeline provides stronger external transfer.

2.5 Datasets for 3D Object Recognition: APS, YCB-28 and ModelNet40

The APS data used in the reference work were generated from artificial box, cylinder and sphere models. The clean set contains 591 point clouds, and an error-corrupted counterpart introduces controlled coordinate perturbations [9,11]. YCB provides standardised everyday objects for manipulation research [13]. The YCB-28 protocol selects 28 objects that can be approximated by the three primitive classes and is therefore the primary real-object test in this project [9,13].

ModelNet40 originates from the ModelNet/3D ShapeNets CAD collection [14]. Because the present classifier has only three outputs, the full 40-class benchmark is not directly compatible. A manually mapped primitive-like subset is therefore used only as an additional external CAD evaluation. This distinction is important: the reported ModelNet result measures transfer on the selected subset, not performance on the official 40-class task.

2.6 Optimisation Algorithms: AdamW and RProp

Neural-network training adjusts weights using gradients of a loss function. Adam combines first-order gradient information with adaptive estimates of the first and second moments of the gradients [15]. AdamW modifies this approach by decoupling weight decay from the gradient-based loss update, providing a clearer form of parameter regularisation [16]. AdamW was used for the main PointNet++ experiments because it is well suited to stochastic mini-batch training and allowed weight decay to be applied directly.

RProp is also a backpropagation learning algorithm, but its parameter updates depend primarily on the sign of the partial derivative rather than the gradient magnitude. Individual step sizes are increased or decreased according to successive gradient signs [17]. This behaviour provided a substantially different optimisation mechanism for the supplementary comparison. However, classical RProp was designed around deterministic or batch gradients, so its behaviour in noisy mini-batch deep learning can differ from adaptive-moment methods. The comparison in this report therefore evaluates its empirical suitability for this particular PointNet++ transfer problem rather than claiming that one optimiser is universally superior.

2.7 Summary of Research Gap

Previous research establishes direct point-cloud learning, hierarchical PointNet++ features, the challenge of synthetic-to-real transfer and the potential value of perturbed APS training. A narrower gap remains in understanding how a PointNet++ classifier trained only on artificial primitive shapes transfers to real-object point clouds when different input representations and synthetic training conditions are used. This project addresses that gap using YCB-28 as the principal real-object evaluation and adds two further checks: transfer to an external ModelNet40 CAD subset and comparison with an alternative optimiser.

3. Methodology

3.1 Overall Experimental Workflow

Figure 1 summarises the experimental workflow. APS data were used for model training, while separately generated APS-style validation data were used for checkpoint selection. YCB-28 and the ModelNet40 primitive-like subset remained external to training and model selection. The first stage established a PointNet++ baseline using XYZ coordinates plus surface normals. The later pipeline used XYZ coordinates only together with the final AdamW training protocol. Clean, error and mixed APS training were each repeated nine times. The mixed AdamW checkpoints were then used for detailed class-wise and object-wise YCB-28 analysis. ModelNet40 provided an additional external CAD evaluation, while RProp was tested separately on the mixed APS condition across ten independent runs.

The implementation and supporting notebooks are available in the project repository.¹

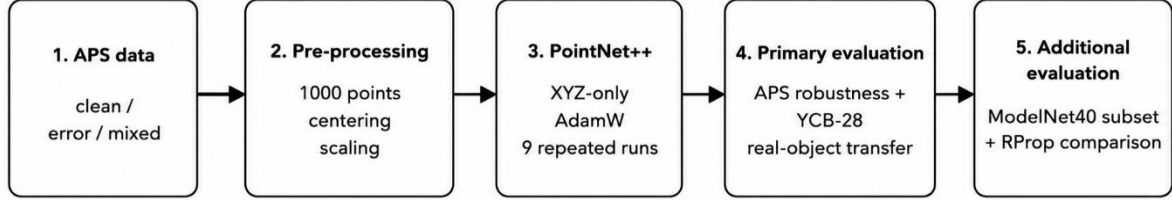


Figure 1: Overall experimental workflow. APS data are used for training and validation, YCB-28 and the ModelNet40 primitive-like subset are external evaluations.

Table 1: Dataset roles and sample counts used in the project.

Dataset	Role	Samples	Classes / mapping
APS-clean	Training	591	220 box, 190 cylinder, 181 sphere
APS-error	Training	591	Noisy counterpart of APS-clean
APS mixed	Training	1182	APS-clean + APS-error
APS-style test	Test	600	200 per primitive class
APS-style validation	Validation	600	200 per primitive class
YCB-28	External real-object test	560	28 objects, 9 box, 8 cylinder, 11 sphere
ModelNet40 subset	External CAD test	486	226 box, 240 cylinder, 20 bowl proxy

3.2 APS Dataset Preparation: Clean, Error and Mixed Sets

The APS-clean training set contained 591 samples: 220 boxes, 190 cylinders and 181 spheres. APS-error contained the same 591 shapes with point-coordinate perturbations. In the reference dataset, each point was displaced within $\delta \in [-0.05L, +0.05L]$, where L is the bounding-box side length, giving a 5% error level [10]. The mixed condition combined APS-clean and APS

¹ <https://github.com/khosekunal98-droid/pointnetpp-primitive-shape-generalisation>

error, giving 1182 training samples. Earlier work showed that perturbed APS shapes can improve robustness to less ideal geometry [9,10].

Separate balanced validation and test sets of 600 samples each were generated, with 200 samples per class. From the clean test set, APS-error used uniform XYZ noise in $[-0.025, +0.025]$, while APS-error-double used $[-0.050, +0.050]$ as a stronger robustness test. These evaluation sets were generated separately from the original training files and were not used for model training or checkpoint selection.

3.3 YCB-28 Real-Object Test Set

The YCB-28 protocol followed the object mapping used in the reference primitive-recognition study [9]. Nine objects were treated as box-like, eight as cylinder-like and eleven as sphere-like. Twenty point clouds were prepared per object, giving 560 test samples. Each sample contained 1000 points. Object meshes were sampled and normalised to the same input convention as the APS data.

YCB-28 was evaluated only after model training and checkpoint selection had been completed. Performance on this set therefore measures synthetic-to-real transfer rather than accuracy after domain-specific fine-tuning. The reported 89.19% mean accuracy is consequently an external test result and was not obtained by training or selecting models using YCB labels.

3.4 ModelNet40 Primitive-Like Subset

The standard point-cloud version of ModelNet40 contains 2048 XYZ points per sample. Because the trained classifier has only three outputs, evaluation was restricted to a manually selected 486-sample primitive-like subset rather than the complete 40-class benchmark. The box-like mapping comprised glass box, xbox, wardrobe and night stand (226 samples). The cylinder-like mapping comprised bottle, cup, vase and flower pot (240 samples). Bowl supplied 20 samples as a weak sphere-like proxy. Each selected sample was centred, scaled using its bounding-box diagonal and reduced to 1000 points. Clean, error and mixed checkpoints were first compared using single-pass evaluation.

After mixed training had been selected as the strongest condition, ten-vote test-time evaluation was applied to the mixed checkpoints only. Each object was evaluated using ten randomly sampled 1000-point subsets, the predicted class probabilities were averaged, and the final class was selected from the averaged probabilities. This procedure changed only the evaluation method and did not retrain the network.

3.5 Point-Cloud Preprocessing and Normalisation

All final models used 1000 XYZ points per sample. In the main AdamW APS-to-YCB pipeline, each point cloud was centred by subtracting its centroid and then scaled by the maximum Euclidean distance from that centroid:

$$p'_i = \frac{p_i - c}{\max_j \|p_j - c\|_2} \quad (1)$$

where (p_i) is the original coordinate vector of point i , (c) is the point-cloud centroid, and (p'_i) is the normalised coordinate vector. This centres the object at the origin and scales the furthest point to unit distance. YCB-28 point-cloud files were first generated from object meshes using bounding-box-diagonal scaling and centring; before inference they were passed through the same AdamW loader, which applied Equation (1). ModelNet40 and the supplementary RProp experiment used centring and bounding-box-diagonal scaling directly. Only XYZ coordinates were retained for the final classifier.

3.6 PointNet++ Classifier Implementation

The classifier was implemented in PyTorch using PointNet++ set abstraction. The final input tensor contained 1000 XYZ points with no normal channel. The first set-abstraction layer sampled 128 centroids using a radius of 0.30 and up to 32 neighbouring points, followed by shared MLP widths of 64-64-128. The second set-abstraction layer sampled 32 centroids using a radius of 0.60 and up to 64 neighbouring points, followed by MLP widths of 128-128-256. A final global set-abstraction stage used MLP widths of 256-512-1024. The resulting 1024-dimensional global feature was passed through fully connected layers of 512 and 256 units before the final three-class output. Batch normalisation, ReLU activations and dropout of 0.4 were used in the classifier.

This hierarchy follows the PointNet++ principle of progressively aggregating local neighbourhood features into a global representation [2]. The model differs from the original PointNet used by Zheng and Castellani [9]; consequently, this study is an extension of their experimental logic rather than an exact reproduction.

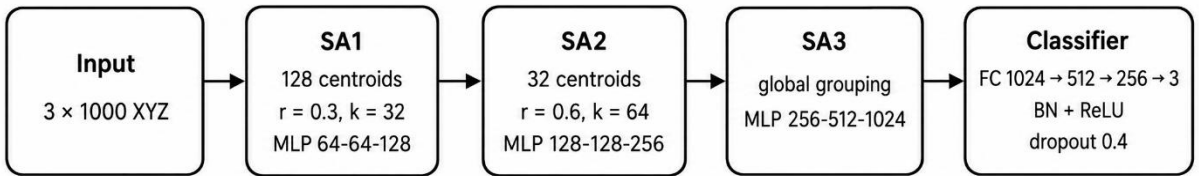


Figure 2: PointNet++ architecture implemented in the final XYZ-only classifier. SA denotes set abstraction.

3.7 AdamW Training Protocol

The final AdamW experiments used a batch size of 16, learning rate 0.0008, weight decay of 1×10^{-4} , and cross-entropy loss with label smoothing 0.05. A weighted sampler and class-weighted loss were used to reduce class imbalance. Clean, error and mixed training were each repeated for nine independent runs, with a maximum of 50 epochs and early-stopping patience of 12. The best checkpoint was selected using APS validation accuracy, with validation loss used to break ties.

Training augmentation included random three-axis rotation, scaling from 0.8 to 1.2, translation from -0.02 to +0.02, Gaussian jitter with standard deviation 0.01 clipped to ± 0.03 , and 5% point dropout. A ReduceLROnPlateau scheduler reduced the learning rate by a factor of 0.5 after five epochs without validation-loss improvement, and gradient clipping used a maximum norm of 1.0. YCB-28 was used only for final external evaluation and not for training or checkpoint selection.

The same AdamW settings were used across clean, error and mixed training so that performance differences mainly reflected the training data rather than changes in optimisation. Nine independent runs per condition reduced reliance on a single random initialisation and enabled repeated-run statistical comparison.

Table 2: Final training settings for the AdamW and RProp experiments.

Setting	AdamW Main experiment	RProp comparison
Input	1000 XYZ points	1000 XYZ points
Training condition	Clean, error and mixed	Mixed only
Independent runs	9 per condition	10
Initial learning rate	0.0008	0.003
Maximum epochs	50	80
Weight decay	1×10^{-4}	-
Label smoothing	0.05	0.0
Sampling / loss balance	Weighted sampler + class-weighted loss	Class-weighted loss
Augmentation	3-axis rotation, scale 0.8-1.2, shift ± 0.02 , jitter $\sigma=0.01$, clip ± 0.03 , 5% point dropout	Mild augmentation

3.8 RProp Optimiser Comparison Protocol

The supplementary RProp experiment retained the XYZ-only PointNet++ architecture and mixed APS training condition. Ten independent runs were trained using a batch size of 16, an initial learning rate of 0.003, a maximum of 80 epochs and early-stopping patience of 20. Mild augmentation consisted of random Z-axis rotation, scaling from 0.95 to 1.05, translation from -0.01 to +0.01 on each axis, and Gaussian jitter with standard deviation 0.002 clipped to ± 0.01 . No point dropout was used.

The weighted sampler and label smoothing used in the main AdamW pipeline were removed, while class-weighted cross-entropy was retained. The experiment was used as a supplementary learning-algorithm comparison rather than as a replacement for the main AdamW training study.

3.9 Evaluation Metrics and Statistical Tests

Overall classification accuracy was the primary metric. Repeated-run results are reported using mean, standard deviation, median, quartiles, minimum and maximum where appropriate. Class-wise accuracy and confusion matrices were used to identify systematic errors, while YCB object-wise accuracy was used to identify individual geometries that violated the ideal primitive abstraction.

Because neural-network learning is stochastic and the number of runs is small, pairwise Mann-Whitney U tests were used instead of assuming normality [18]. A Kruskal–Wallis test provided an omnibus comparison of the three AdamW conditions [19]. For the three pairwise tests, Holm's sequential correction controlled family-wise error [20]. Statistical significance was assessed at $\alpha=0.05$. The optimiser comparison used a two-sided Mann-Whitney U test between the nine mixed AdamW accuracies and ten RProp accuracies. Statistical testing supports the repeated-run comparisons but does not remove dataset-design limitations.

4. Results

4.1 APS Test Results

The final XYZ-only models retained very high accuracy on controlled APS-style data. Table 3 shows that clean training was effectively perfect on APS-clean and APS-error but deteriorated sharply on the stronger APS-error-double test, where mean accuracy fell to 67.52% with high run-to-run variation. In contrast, error and mixed training remained above 98.8% on APS-error-double while also retaining approximately 100% accuracy on clean APS. These results indicate that exposure to modified geometry substantially improved robustness to stronger perturbations.

Training and validation behaviour was examined across all 27 final AdamW runs. No train–validation accuracy gap exceeded 5 percentage points. Validation accuracy was sometimes slightly higher than training accuracy because augmentation was applied only during training. The learning curves therefore showed no clear evidence of substantial overfitting within the APS experiments.

Table 3: XYZ-only PointNet++ accuracy (%) on APS-style test sets across nine AdamW runs.

Training condition	APS-clean mean $\pm$ SD	APS-error mean $\pm$ SD	APS-error-double mean $\pm$ SD
Clean	100.00 $\pm$ 0.00	99.83 $\pm$ 0.12	67.52 $\pm$ 12.49
Error	99.96 $\pm$ 0.07	99.91 $\pm$ 0.12	98.87 $\pm$ 0.91
Mixed	99.98 $\pm$ 0.06	99.93 $\pm$ 0.17	98.89 $\pm$ 0.83

4.2 XYZ+Normals Baseline versus Final XYZ-Only Pipeline

The first PointNet++ evaluation used XYZ coordinates plus surface normals. Although performance on APS data was high, transfer to YCB-28 was substantially weaker: the clean, error and mixed training conditions achieved mean classification accuracies of 66.05%, 67.32% and 67.18%, respectively. The later XYZ-only pipeline achieved corresponding means of 83.06%, 86.55% and 89.19%. For the mixed condition, the difference between the two pipelines was approximately 22 percentage points.

The later XYZ-only experiments also used the final, stronger AdamW training protocol. The improvement should therefore not be interpreted as a controlled ablation proving that removal of normals alone caused the increase. Instead, the comparison shows that the complete XYZ-only pipeline transferred substantially better to YCB-28 and was therefore retained for the final experiments. The mixed XYZ-only condition also showed relatively low run-to-run variation, with a standard deviation of 1.90 percentage points.

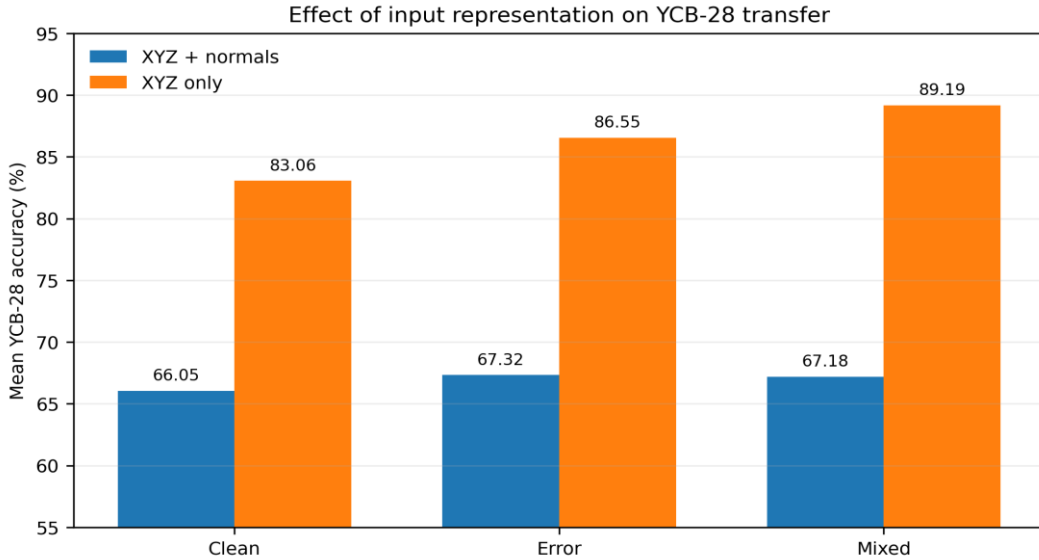


Figure 3: Mean classification accuracy on YCB-28 for the XYZ-plus-normal baseline & final XYZ-only pipeline.

Table 4: Descriptive statistics for classification accuracy on the YCB-28 test set across repeated runs (%).

Input	Training	Min	Q1	Median	Q3	Max	Mean	SD
XYZ+N	Clean	61.96	64.11	64.46	65.54	75.00	66.05	4.19
XYZ+N	Error	58.39	64.29	67.14	70.18	73.39	67.32	4.58
XYZ+N	Mixed	63.39	65.00	68.96	69.82	72.14	67.18	3.15
XYZ	Clean	80.71	81.61	82.68	84.46	86.07	83.06	1.90
XYZ	Error	83.04	84.82	86.61	87.14	90.36	86.55	2.41
XYZ	Mixed	85.71	87.68	90.00	90.54	91.25	89.19	1.90

4.3 YCB-28 Main AdamW Results

Table 4 summarises classification accuracy on YCB-28 across the nine XYZ-only runs for each training condition. Clean training achieved a mean of $83.06 \pm 1.90\%$, error training achieved $86.55 \pm 2.41\%$, and mixed training achieved $89.19 \pm 1.90\%$. The mixed condition also had a median of 90.00% and a best run of 91.25% , with all nine runs between 85.71% and 91.25% . Figure 4 shows that the mixed distribution is shifted upward relative to the clean and error conditions rather than being driven by a single exceptional run. Mixed training therefore provided the strongest repeated-run performance on YCB-28 and was selected as the final AdamW training condition.

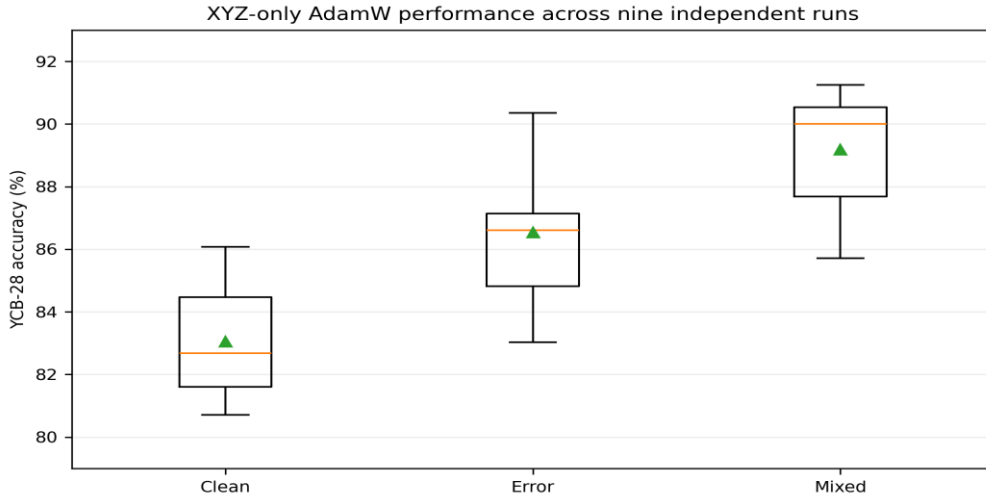


Figure 4: Distribution of YCB-28 accuracy for the three final XYZ-only AdamW training conditions.

4.4 YCB-28 Class-Wise and Object-Wise Findings

The final mixed model was balanced across the three primitive classes: box 88.70% , cylinder 87.36% and sphere 90.91% . The aggregated confusion matrix shows that most residual errors were between boxes and cylinders, while spheres were the most reliable class. Across nine runs, 1,800 of 1,980 sphere predictions were correct.

At object level, 25 of 28 YCB objects were majority-correct and 14 achieved 100% accuracy across the aggregated predictions. The three weakest objects were 063_a_marbles (0%), 040_large_marker (32.22%) and 026_sponge (35%). These are boundary cases for the primitive abstraction: the marbles form a compound cluster rather than one sphere, the marker is an unusually thin cylinder, and the sponge is a rounded irregular cuboid. These failures therefore reflect limitations of the three-class geometric mapping rather than a general failure of one primitive class.

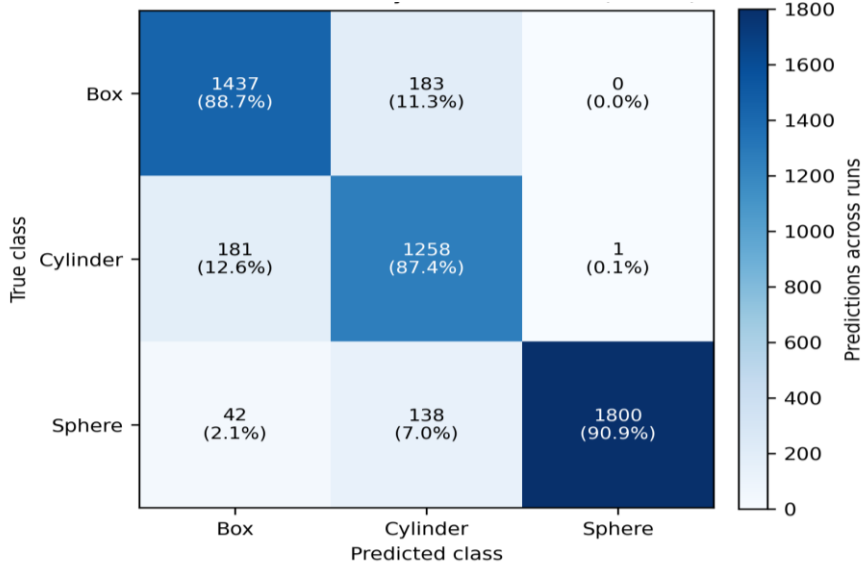


Figure 5: Aggregated YCB-28 confusion matrix for the final mixed XYZ-only AdamW model across nine runs. Percentages are normalised by true class.

Table 5: Class-wise accuracy & principal object-level failure cases for final mixed AdamW model on YCB-28.

Class / item	Accuracy / result	Interpretation
Box class	88.70%	Strong transfer, remaining errors mainly box→cylinder
Cylinder class	87.36%	Strong but most ambiguous primitive group
Sphere class	90.91%	Highest class-wise accuracy
063_a_marbles	0.00%	Compound cluster, not a single sphere
040_large_marker	32.22%	Very thin elongated cylinder
026_sponge	35.00%	Irregular rounded cuboid

4.5 Statistical Comparison of Clean, Error and Mixed Training

The Kruskal–Wallis test found a significant overall difference among the three XYZ-only AdamW training conditions when evaluated on YCB-28 ($H=16.6427$, $p=0.000243$). Pairwise Mann–Whitney U tests remained significant after Holm correction: clean versus error, $U=9.0$, adjusted $p=0.0123$; clean versus mixed, $U=1.0$, adjusted $p=0.00172$; and error versus mixed, $U=14.0$, adjusted $p=0.0213$. The repeated-run statistical analysis therefore supports the ordering observed in the descriptive results, with mixed training producing the highest YCB-28 accuracy.

Table 6: Pairwise Mann-Whitney U tests comparing YCB-28 classification accuracy across the XYZ-only AdamW training conditions. Omnibus Kruskal–Wallis: $H=16.6427$, $p=0.000243$.

Comparison	U	Raw p	Holm-adjusted p	Significant (0.05)
Clean vs Error	9.0	0.006167	0.012334	Yes
Clean vs Mixed	1.0	0.000574	0.001721	Yes
Error vs Mixed	14.0	0.021346	0.021346	Yes

4.6 ModelNet40 Primitive-Like Subset Results

The ModelNet40 evaluation used 486 samples from nine manually selected categories: 226 box-like samples, 240 cylinder-like samples and 20 bowl samples used as a weak sphere-like proxy. In single-pass testing, the clean, error and mixed checkpoints achieved mean accuracies of 71.58%, 72.57% and 79.81%, respectively. The overall difference was significant (Kruskal–Wallis $H=10.0669$, $p=0.0065$). Holm-adjusted Mann–Whitney tests showed that mixed training performed significantly better than clean training ($p=0.0078$) and error training ($p=0.0424$), while clean and error training were not significantly different ($p=0.7908$).

Mixed training was therefore selected for ten-vote evaluation. This increased mean accuracy from 79.81% to 81.00% and the best result from 83.54% to 84.98%. Box-like accuracy was 96.07%, while cylinder-like accuracy was 73.56%. The bowl proxy achieved 0% accuracy; across 180 bowl predictions, 171 were classified as cylinders and none as spheres. Because ModelNet40 contains no true sphere or ball category, this result should be interpreted primarily as a limitation of the manual mapping rather than evidence of poor performance on genuine spherical objects. The reported values apply only to the selected primitive-like subset and not to the full ModelNet40 benchmark.

Table 7: ModelNet40 primitive-like subset mapping. Bowl is a weak proxy rather than a true sphere.

Mapped primitive	ModelNet40 categories	Samples
Box-like	glass_box, xbox, wardrobe, night_stand	226
Cylinder-like	bottle, cup, vase, flower_pot	240
Sphere-like proxy	bowl	20

Table 8: ModelNet40 primitive-like subset results. These are not full 40-class benchmark results.

Evaluation	Runs	Mean (%)	Median (%)	Best (%)	Main note
Clean single-pass	9	71.58	73.87	76.34	Baseline clean-trained condition
Error single-pass	9	72.57	73.25	81.89	Error trained condition
Mixed single-pass	9	79.81	82.10	83.54	Best single-pass condition
Mixed 10-vote	9	81.00	82.30	84.98	+1.19 pp mean vs single-pass

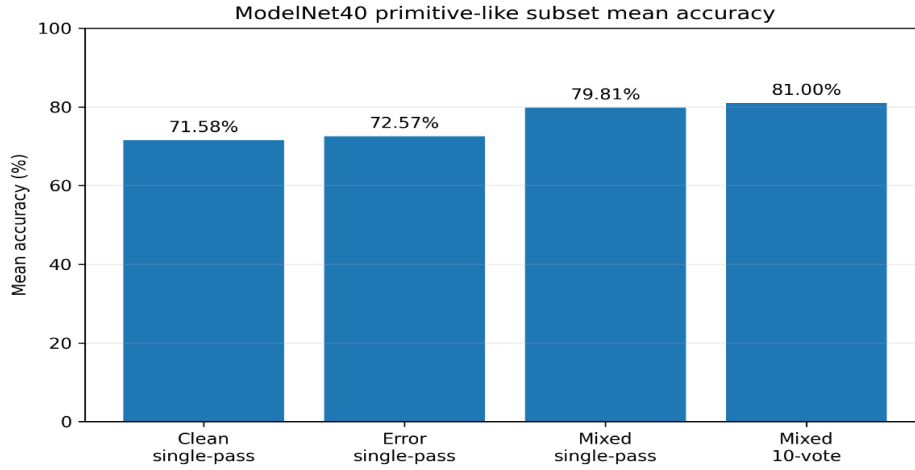


Figure 6: ModelNet40 primitive-like subset mean accuracy. Ten-vote testing was applied only after mixed training had been selected by the single-pass comparison.

4.7 RProp Optimiser Comparison

RProp transferred substantially less well to YCB-28 than the final AdamW configuration. Across ten runs, accuracy ranged from 68.75% to 72.86%, with a mean of 71.20%, median of 71.61% and standard deviation of 1.35%. The mixed AdamW model achieved $89.19 \pm 1.90\%$, a mean advantage of 17.99 percentage points. A two-sided Mann–Whitney U test confirmed a clear difference between the repeated-run distributions ($U=90.0$, $p=0.000276$; Figure 7).

RProp class-wise accuracies were 84.94% for boxes, 29.81% for cylinders and 90.05% for spheres, making cylinder recognition its main weakness. Because the AdamW and RProp experiments also differed in augmentation, weighted sampling and label smoothing, this should be interpreted as a comparison between the two final training configurations rather than a pure optimiser ablation. AdamW was therefore retained for the final model.

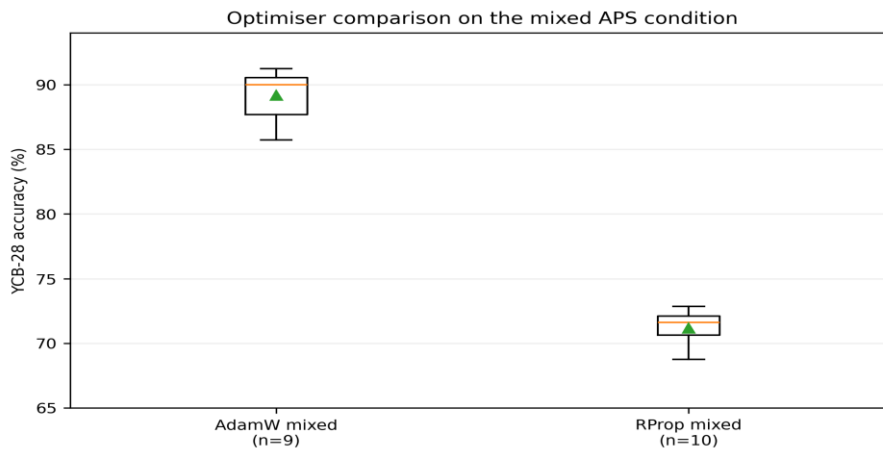


Figure 7: YCB-28 repeated-run classification accuracy for the final mixed AdamW model and the supplementary mixed RProp model.

Table 9: AdamW versus RProp on the mixed APS training condition, evaluated on YCB-28.

Optimiser	Runs	Mean (%)	Median (%)	Best (%)	SD (%)	Mann–Whitney result
AdamW	9	89.19	90.00	91.25	1.90	U=90.0, p=0.000276
RProp	10	71.20	71.61	72.86	1.35	AdamW > RProp

4.8 Summary of Final Results

The final configuration used mixed APS-clean plus APS-error training with XYZ-only PointNet++ and AdamW optimisation. It achieved 89.19% mean classification accuracy on YCB-28, retained 98.89% mean accuracy on the APS-error-double robustness test, and produced the strongest performance among the three training conditions on the ModelNet40 primitive-like subset. The RProp configuration did not improve external transfer.

5. Discussion

5.1 Comparison with Zheng and Castellani’s PointNet Results

The closest published comparison is Zheng and Castellani’s APS-to-YCB study using original PointNet [9]. Across ten learning trials, their reported YCB-28 median accuracies were 78.84% for APS-clean, 85.98% for APS-error and 80.71% for YCB-similar training. Their later optimiser study reported further PointNet experiments with Adam and CEST. Optimised CEST reached median YCB accuracies of 89.20% on APS-error and 88.21% on APS-all [10].

The present mixed-trained XYZ-only PointNet++ model achieved a median classification accuracy of 90.00% and a mean of 89.19% on YCB-28. These values are competitive with the strongest published APS-to-YCB results, but they do not constitute a controlled architecture-only comparison because the PointNet++ architecture, optimiser, augmentation, training composition and preprocessing differ from the reference studies. The appropriate conclusion is therefore that PointNet++ successfully extends the earlier synthetic-to-real experimental framework and that the selected mixed-training pipeline produced strong external transfer. The results do not establish that PointNet++ alone is superior to PointNet or CEST.

Table 10: Contextual comparison with the reference PointNet study. Values are not a controlled architecture-only benchmark because training procedures differ.

Study / model	Training condition	YCB-28 statistic	Accuracy (%)
Zheng and Castellani PointNet [9]	APS-clean	Median (10 runs)	78.84
Zheng and Castellani PointNet [9]	APS-error	Median (10 runs)	85.98
Zheng and Castellani PointNet [9]	YCB-similar	Median (10 runs)	80.71
This study PointNet++	APS-clean, XYZ-only	Median (9 runs)	82.68
This study PointNet++	APS-error, XYZ-only	Median (9 runs)	86.61
This study PointNet++	Mixed, XYZ-only	Median (9 runs)	90.00

5.2 Interpretation of the XYZ-Only Transfer Improvement

The external-transfer comparison suggests that the earlier pipeline may have relied on surface-normal information that did not transfer consistently between APS and YCB. Surface normals encode local orientation and are theoretically useful for distinguishing planar and curved surfaces, but their values can depend on mesh construction, sampling density and the normal-estimation convention. XYZ-only input removes this additional feature channel, while PointNet++ can still learn local geometric structure directly from spatial neighbourhoods.

The mixed-condition mean on YCB-28 increased by approximately 22 percentage points between the earlier XYZ-plus-normal baseline and the final XYZ-only pipeline. However, the later experiments also used a stronger training protocol. The improvement therefore cannot be attributed to normal removal alone. A controlled future ablation should calculate normals using the same estimator and neighbourhood settings for every dataset and change only the input representation.

5.3 Why Mixed Training Was Best

Clean APS provides ideal examples of each primitive class, whereas APS-error introduces geometric variation around those ideal shapes. Mixed training combines both types of examples and may therefore preserve the main class geometry while exposing the classifier to a wider range of deviations. This interpretation is consistent with the results. On APS-error-double, clean training fell to 67.52% mean accuracy, whereas mixed training retained 98.89%. On YCB-28, the means increased from 83.06% for clean training to 86.55% for error training and 89.19% for mixed training. The pairwise statistical tests showed significant differences among all three conditions.

The result also aligns qualitatively with Zheng and Castellani's finding that error-corrupted APS improved PointNet transfer [9]. The present study extends that observation by showing that combining clean and error examples was stronger than either condition alone for the final PointNet++ pipeline.

5.4 ModelNet40 Subset Interpretation and Limitations

ModelNet40 is useful because its CAD objects constitute a different external source from both APS and YCB. Strong box accuracy and moderate cylinder accuracy therefore provide additional evidence that the mixed model learned transferable shape cues. Nevertheless, the subset is curated and its classes were manually mapped. It must not be described as an 81% result on the full ModelNet40 benchmark.

The bowl failure illustrates why semantic categories and primitive classes cannot be equated automatically. A bowl contains substantial curved surface area, but its opening and concavity make its global topology closer to a hollow cylindrical shell than to a closed sphere. The absence of true sphere categories means ModelNet40 provides essentially no fair test of the

sphere output. YCB-28, which contains balls and fruit-like objects, remains the appropriate sphere evaluation in this project.

YCB-28 therefore remains the stronger three-class external evaluation because it contains real everyday objects that can reasonably be mapped to all three target primitive classes. ModelNet40 serves a complementary purpose by testing transfer to a separate CAD dataset with different semantic categories and point-cloud sampling. Agreement between the two evaluations provides additional evidence for box and cylinder transfer, but the ModelNet40 result should not be given the same weight as YCB-28 when judging overall three-class performance because it lacks a valid sphere category.

5.5 Interpretation of the AdamW-RProp Comparison

AdamW uses gradient magnitude together with adaptive estimates of the first and second moments of the gradients and is designed for stochastic mini-batch optimisation [15,16]. RProp instead adapts parameter-specific step sizes mainly from changes in gradient sign [17]. In a point-cloud training pipeline containing random augmentation, mini-batch sampling and varied geometry, gradient signs can fluctuate between updates. This provides one possible explanation for the weaker transfer observed with the RProp configuration.

The RProp models trained successfully on the mixed APS condition, but their mean YCB-28 accuracy remained at 71.20%. The strongest class was sphere at 90.05%, while cylinder accuracy fell to 29.81%. Real YCB cylinder-like objects include cans, cups, a mug, a marker and pitcher-like geometry, making this class geometrically more varied than the synthetic APS cylinders. Nevertheless, the two experiments also used different augmentation and class-balancing settings. The result therefore does not establish a general theoretical advantage of AdamW over RProp; it shows that the final AdamW training configuration was substantially more effective for the transfer problem examined here.

5.6 Limitations to Validity

Several limitations constrain generalisation. Only three primitive classes were considered, and APS is a relatively small synthetic dataset. YCB-28 contains repeated samples from 28 object identities, so high sample accuracy does not guarantee equivalent performance on arbitrary unseen object families. The ModelNet40 subset is manually curated and its bowl mapping is not a valid sphere benchmark. Hyperparameter exploration was practical rather than exhaustive, and RProp was evaluated only for mixed training. Finally, no sensor-native or end-to-end robot experiment was conducted. The reported results therefore establish classification behaviour under the defined protocols, while the proposed benefits for grasping and disassembly remain to be validated at task level.

6. Impact Statement

6.1 Industrial and Robotics Relevance

Primitive-shape recognition could support robotic perception in disassembly, remanufacturing and automated handling by narrowing candidate grasp positions, end-effector orientations and downstream handling strategies. Training mainly on synthetic APS data may also reduce the amount of labelled real 3D data required during early development. For manufacturers and robotic-system integrators, this could provide a lower-cost route for developing and testing perception systems before larger real-world trials. The three-class output could also serve as an initial geometric filtering stage before more specialised pose estimation or grasp planning. However, any commercial benefit would depend on sensing hardware, computing requirements, system integration and task-level accuracy, and this project did not demonstrate direct production or cost savings.

6.2 Environmental and Societal Impact

Improved robotic disassembly and remanufacturing could support the reuse, repair and recovery of components that might otherwise be discarded. More reliable geometric recognition may help robots sort and handle parts more effectively, supporting material recovery and waste reduction. Automated perception could also reduce some repetitive manual inspection tasks. These are potential benefits rather than outcomes directly demonstrated in this project, and they must be balanced against the computing and energy required to train and run deep-learning models.

A further consideration is the balance between computational cost and the potential benefits of more efficient robotic handling. Although training deep-learning models consumes energy, the relatively small three-class classification task used here is less demanding than large-scale vision models. If such a system contributed to more reliable sorting, reuse or recovery of components, the environmental benefit could extend beyond the perception task itself. However, these gains would depend on the complete robotic process, including sensing, computation and mechanical operation. Future deployment should therefore consider both recognition performance and the overall energy and resource cost of the system.

6.3 Stakeholders and Implementation Considerations

The main stakeholders include remanufacturing companies, manufacturers, robotic-system integrators, automation engineers and researchers working on 3D perception. Before deployment, the classifier would require testing on sensor-based partial point clouds, a wider range of objects and real operating conditions. Confidence thresholds and methods for handling objects outside the three primitive classes would also be needed. Misclassification could lead to an unsuitable grasp or robot motion, so the classifier should not be used as a stand-alone decision system. It should instead be combined with pose estimation, collision checking and grasp validation before industrial use.

7. Conclusions and Future Work

7.1 Main Conclusions

This project showed that an XYZ-only PointNet++ classifier trained using mixed APS-clean and APS-error data with AdamW can generalise beyond the synthetic APS domain. Across nine independent runs, the final model achieved 89.19% mean classification accuracy on the YCB-28 real-object test set, with class-wise accuracies of 88.70% for boxes, 87.36% for cylinders and 90.91% for spheres. Mixed training also retained 98.89% mean accuracy on the stronger APS-error-double robustness test.

On the ModelNet40 primitive-like subset, the mixed model achieved 81.00% mean accuracy using ten-vote test-time evaluation, although the bowl mapping does not provide a valid benchmark for true spherical objects. The supplementary RProp configuration achieved 71.20% mean accuracy on YCB-28 and did not improve on the final AdamW configuration.

Overall, the strongest external transfer was obtained using mixed synthetic training, XYZ-only input and AdamW optimisation. The results also show that very high performance on clean synthetic point clouds does not by itself guarantee transfer to independent real-object data, and that the choice of external class mapping must be considered carefully when evaluating primitive-shape classifiers.

7.2 Future Work

The most important next step is evaluation using sensor-native partial point clouds, including occlusion, missing regions and measurement noise, because the current external tests use prepared object models rather than direct robot-camera observations. A controlled normal-estimation study could then determine whether surface information can be reintroduced without increasing domain mismatch. Further work could also expand the primitive vocabulary and compare alternative architectures under exactly the same training and evaluation protocol. Finally, integrating the classifier with pose estimation and grasp planning would show whether improved primitive recognition produces measurable benefits in a real robotic disassembly or handling task.

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Appendices

Appendix A. Dataset object and category lists

The following tables summarise the YCB-28 object mapping and the ModelNet40 primitive-like category mapping used in this project.

Appendix Table A1. YCB-28 primitive-class mapping used in this project.

Primitive class	No. of objects	Objects
Box-like	9	cracker box, sugar box, pudding box, gelatin box, potted meat can, sponge, wood block, foam brick, Rubik's cube
Cylinder-like	8	chips can, master chef can, tomato soup can, tuna fish can, pitcher base, mug, large marker, cups
Sphere-like	11	strawberry, lemon, peach, orange, plum, softball, baseball, tennis ball, racquetball, golf ball, marbles

Appendix Table A2. ModelNet40 primitive-like category mapping used in this project.

Mapped primitive	No. of categories	ModelNet40 categories
Box-like	4	glass_box, xbox, wardrobe, night_stand
Cylinder-like	4	bottle, cup, vase, flower_pot
Sphere-like proxy	1	bowl

Appendix B. Hyperparameter summary

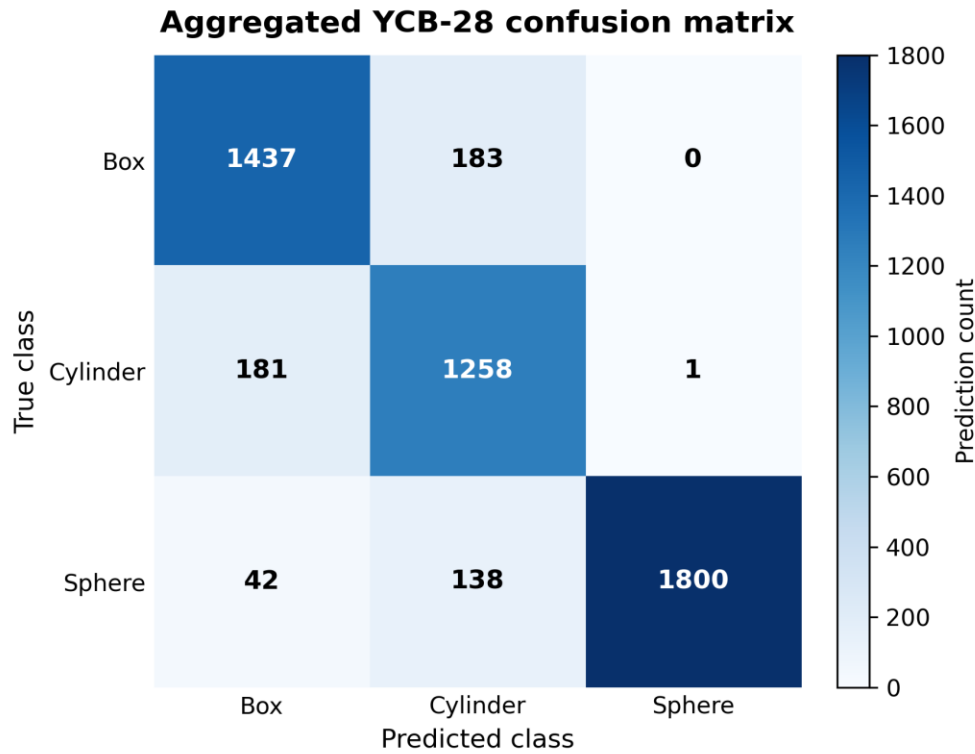
Appendix Table B1 provides a compact comparison of the final AdamW configuration and the supplementary RProp comparison.

Appendix Table B1. Hyperparameter settings for the final AdamW and supplementary RProp experiments.

Setting	AdamW main experiment	RProp comparison
Architecture	PointNet++	PointNet++
Input	XYZ only	XYZ only
Points per sample	1000	1000
APS training condition	Clean, error and mixed	Mixed only
Optimiser	AdamW	RProp
Independent runs	9 per condition	10
Batch size	16	16
Initial learning rate	0.0008	0.003
Weight decay	1×10^{-4}	-
Loss	Cross-entropy	Class-weighted cross-entropy
Label smoothing	0.05	0.0
Maximum epochs	50	80
Early-stopping patience	12	20
Weighted sampler	Yes	No
Augmentation	Rotation, scale, jitter, shift and point dropout	Mild augmentation

Appendix C. Additional confusion-matrix details

The aggregated confusion matrix below shows the exact prediction counts for the final mixed AdamW model across nine independent runs. Rows represent true classes and columns represent predicted classes.



Appendix Figure C1. Aggregated YCB-28 confusion matrix for the final mixed AdamW model across nine runs.

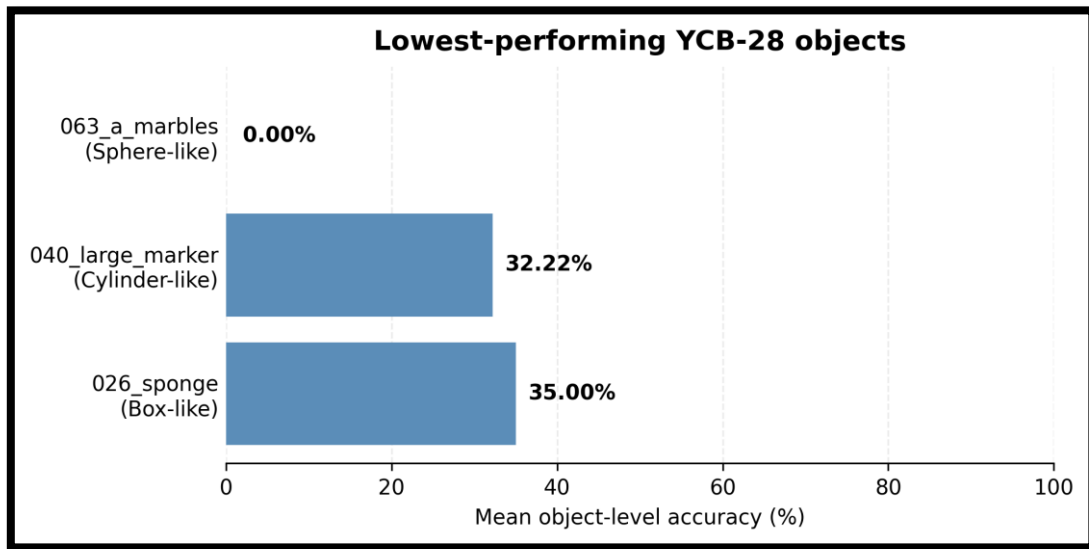
The confusion matrix aggregates predictions across the nine independently trained mixed AdamW checkpoints. Consequently, the counts represent repeated evaluation predictions rather than unique physical objects.

Appendix D. Object-wise result notes

The three YCB-28 objects with mean object-level accuracy below 50% are summarised in Appendix Table D1. The accompanying figure provides a visual comparison of these lowest-performing cases.

Appendix Table D1. Lowest-performing YCB-28 objects for the final mixed AdamW model.

Object	Primitive class	Mean accuracy (%)	Main interpretation
marbles	Sphere-like	0.00	Compound group of marbles rather than a single spherical object
large marker	Cylinder-like	32.22	Thin and highly elongated cylindrical geometry
sponge	Box-like	35.00	Irregular, rounded cuboid geometry



Appendix Figure D1. Mean object-level accuracy of the three lowest-performing YCB-28 objects.

Twenty-five of the 28 YCB objects were majority-correct, while 14 objects achieved 100% object-level accuracy across the aggregated repeated predictions. Full checkpoint-level prediction records are retained in the project results files.

Appendix E. Detailed statistical test outputs

Appendix Table E1. Detailed non-parametric test outputs used in the report.

Test / comparison	Statistic	Raw p	Adjusted p
Clean vs Error	9.0	0.006166754	0.012333508
Clean vs Mixed	1.0	0.000573634	0.001720901
Error vs Mixed	14.0	0.021346320	0.021346320
ModelNet40 Kruskal–Wallis clean/error/mixed	H=10.0669	0.0065	-
AdamW mixed vs RProp	U=90.0	0.000276259	-
ModelNet clean vs error	U=37.0	-	0.7908
ModelNet clean vs mixed	U=6.0	-	0.0078
ModelNet error vs mixed	U=14.0	-	0.0424

Appendix F. Abbreviations and Symbols

The abbreviations and mathematical/statistical symbols used in this report are defined below.

Appendix Table F1. Technical abbreviations and notation used in this report.

Abbreviation / notation	Definition
AdamW	Adam optimiser with decoupled weight decay
APS	Artificial primitive-shape dataset
CAD	Computer-aided design
CEST	CoEvolutionary Selection Training
FC	Fully connected layer
MLP	Multilayer perceptron
pp	Percentage points
Q1	First quartile
Q3	Third quartile
ReLU	Rectified Linear Unit
RProp	Resilient backpropagation
SA	Set abstraction
SD	Standard deviation
XYZ	Three-dimensional Cartesian coordinates (x), (y) and (z)
XYZ+N	XYZ coordinates with surface-normal features
YCB	Yale-CMU-Berkeley object and model set
BN	Batch normalisation

Appendix Table F2. Mathematical and statistical symbols used in this report.

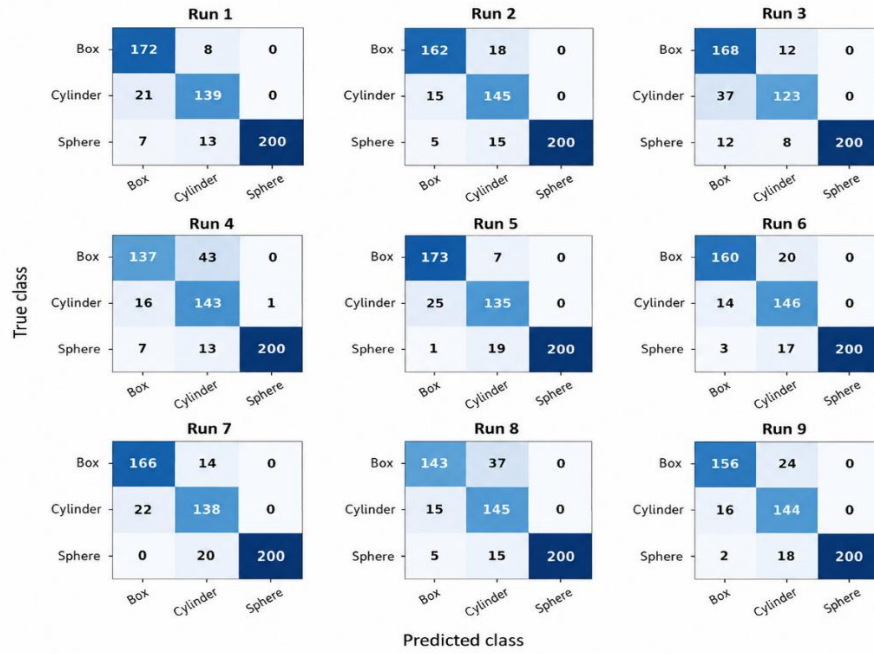
Symbol	Definition
p_i	Original coordinate vector of point (i) in a point cloud
p_i'	Normalised coordinate vector of point (i)
c	Centroid of the point cloud
d_B	Euclidean length of the axis-aligned bounding-box diagonal
H	Kruskal–Wallis test statistic
U	Mann–Whitney U test statistic
p	Statistical p-value
α	Statistical significance level

Appendix G. YCB-28 results

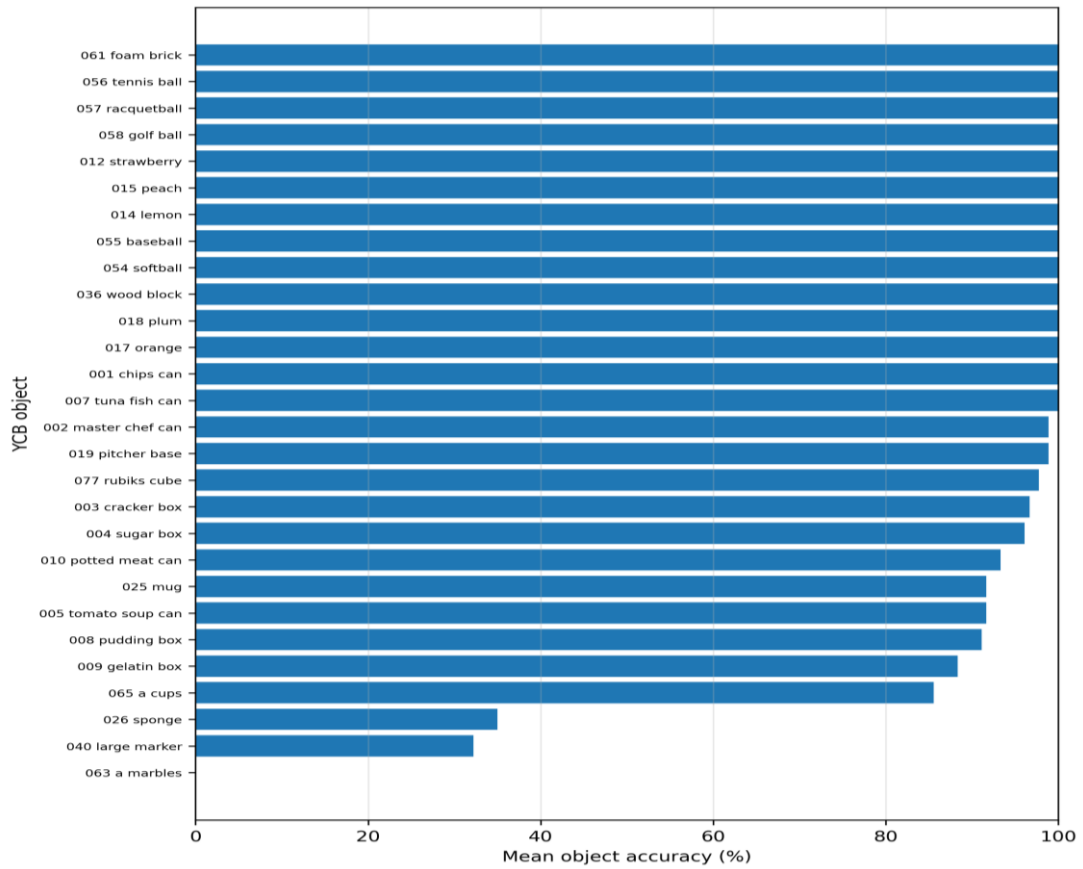
This appendix provides the run-level and object-level evidence behind the YCB-28 summaries reported in the main text. It is included for traceability and does not replace the descriptive statistics, class-wise results or aggregate confusion matrix presented in Section 4.

Appendix Table G1. Run-level YCB-28 accuracy for all final XYZ-only AdamW experiments.

Training condition	Run	Accuracy (%)	Samples
Clean	1	80.89	560
Clean	2	83.57	560
Clean	3	82.32	560
Clean	4	80.71	560
Clean	5	85.18	560
Clean	6	86.07	560
Clean	7	84.46	560
Clean	8	81.61	560
Clean	9	82.68	560
Error	1	86.61	560
Error	2	90.00	560
Error	3	86.61	560
Error	4	84.64	560
Error	5	90.36	560
Error	6	87.14	560
Error	7	85.71	560
Error	8	83.04	560
Error	9	84.82	560
Mixed	1	91.25	560
Mixed	2	90.54	560
Mixed	3	87.68	560
Mixed	4	85.71	560
Mixed	5	90.71	560
Mixed	6	90.36	560
Mixed	7	90.00	560
Mixed	8	87.14	560
Mixed	9	89.29	560



Appendix Figure G1. YCB-28 confusion matrices for each of the nine mixed-trained AdamW checkpoints. Rows are true classes and columns are predicted classes.



Appendix Figure G2. Mean object-wise YCB-28 accuracy for the final mixed-trained AdamW model, averaged across nine independent runs.

Appendix Table G2. Object-wise YCB-28 accuracy for the final mixed-trained AdamW model.

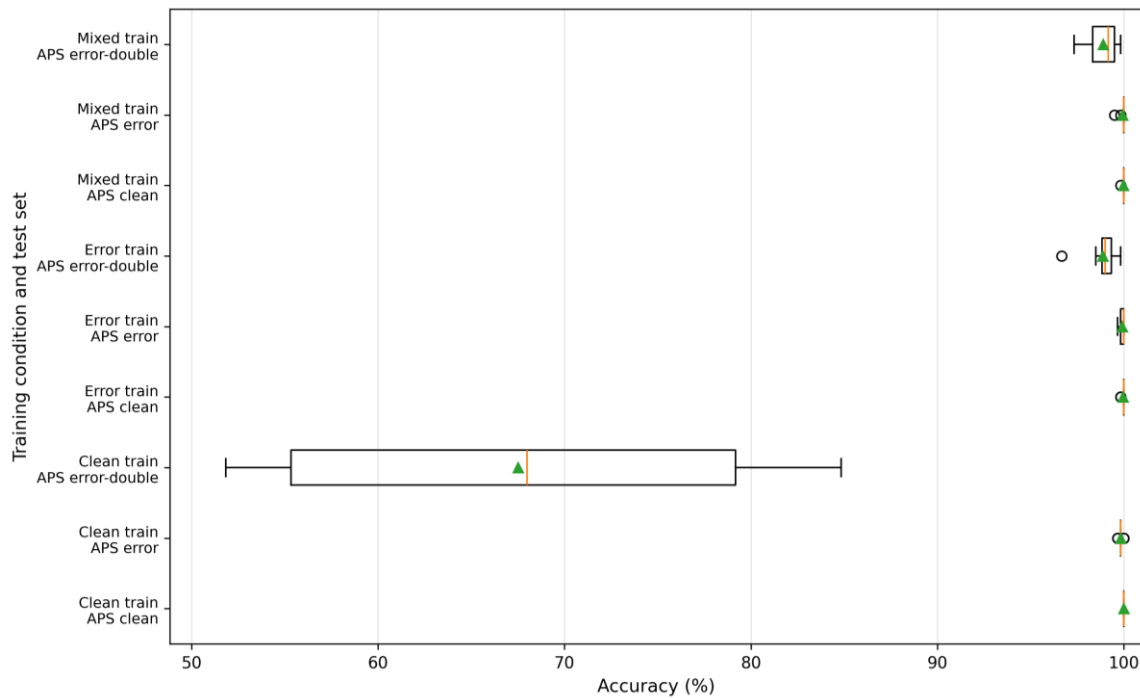
Object	Class	Mean accuracy (%)	SD (%)	Median (%)
063 a marbles	sphere	0.00	0.00	0.00
040 large marker	cylinder	32.22	28.07	30.00
026 sponge	box	35.00	32.21	20.00
065 a cups	cylinder	85.56	11.30	90.00
009 Gelatine box	box	88.33	16.20	95.00
008 pudding box	box	91.11	7.41	90.00
005 tomato soup can	cylinder	91.67	10.00	95.00
025 mug	cylinder	91.67	9.01	95.00
010 potted meat can	box	93.33	6.12	95.00
004 sugar box	box	96.11	6.51	100.00
003 cracker box	box	96.67	7.07	100.00
077 Rubiks cube	box	97.78	3.63	100.00
019 pitcher base	cylinder	98.89	3.33	100.00
002 master chef can	cylinder	98.89	3.33	100.00
007 tuna fish can	cylinder	100.00	0.00	100.00
001 chips can	cylinder	100.00	0.00	100.00
017 orange	sphere	100.00	0.00	100.00
018 plum	sphere	100.00	0.00	100.00
036 wood block	box	100.00	0.00	100.00
054 softball	sphere	100.00	0.00	100.00
055 baseball	sphere	100.00	0.00	100.00
014 lemon	sphere	100.00	0.00	100.00
015 peach	sphere	100.00	0.00	100.00
012 strawberry	sphere	100.00	0.00	100.00
058 golf ball	sphere	100.00	0.00	100.00
057 racquetball	sphere	100.00	0.00	100.00
056 tennis ball	sphere	100.00	0.00	100.00
061 foam brick	box	100.00	0.00	100.00

Appendix H. APS robustness results

The main report summarises APS performance by training condition. The table and box plot below provide the complete descriptive view across the three APS-style test sets and show the pronounced failure of clean-only training under the doubled-error condition.

Appendix Table H1. Detailed APS test accuracy summary across nine runs.

Training	Test set	Mean (%)	SD (%)	Min (%)	Median (%)	Max (%)	Runs
Clean	APS clean	100.00	0.00	100.00	100.00	100.00	9
Clean	APS error	99.83	0.12	99.67	99.83	100.00	9
Clean	APS error-double	67.52	12.49	51.83	68.00	84.83	9
Error	APS clean	99.96	0.07	99.83	100.00	100.00	9
Error	APS error	99.91	0.12	99.67	100.00	100.00	9
Error	APS error-double	98.87	0.91	96.67	99.00	99.83	9
Mixed	APS clean	99.98	0.06	99.83	100.00	100.00	9
Mixed	APS error	99.93	0.17	99.50	100.00	100.00	9
Mixed	APS error-double	98.89	0.83	97.33	99.17	99.83	9



Appendix Figure H1. Distribution of APS test accuracy across repeated runs for each training condition and APS-style test set.

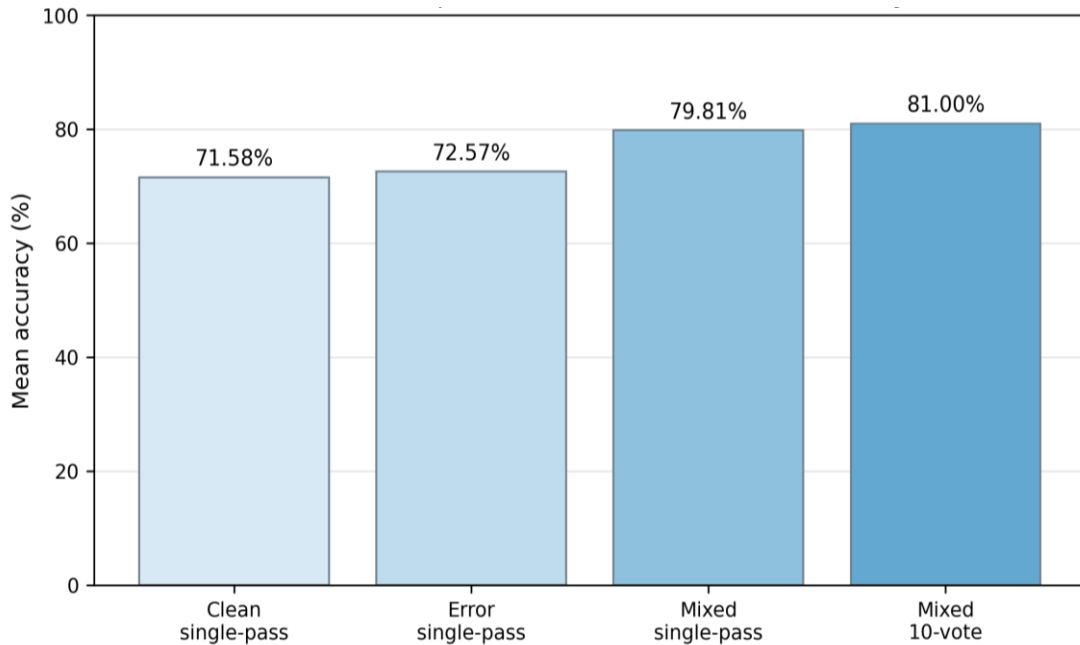
Appendix I. ModelNet40 Primitive-Like Subset Results

ModelNet40 was used as an additional external evaluation through a manually selected primitive-like subset of 486 samples. The full 40-class benchmark was not used because the trained PointNet++ classifier has only three output classes: box, cylinder and sphere. The results should therefore be interpreted only as performance on the selected primitive-like subset.

Appendix Table II. Single-pass ModelNet40 primitive-like subset results.

Training condition	Runs	Mean accuracy (%)	Median (%)	Best (%)
Clean-trained XYZ-only	9	71.58	73.87	76.34
Error-trained XYZ-only	9	72.57	73.25	81.89
Mixed-trained XYZ-only	9	79.81	82.10	83.54

Mixed training produced the highest single-pass mean accuracy and was therefore selected before the voting evaluation. The Kruskal-Wallis test showed an overall difference between the three conditions ($H = 10.0669$, $p = 0.0065$). After Holm correction, mixed training performed significantly better than clean training ($U = 6.0$, $p = 0.0078$) and error training ($U = 14.0$, $p = 0.0424$), while clean and error training were not significantly different ($U = 37.0$, $p = 0.7908$).

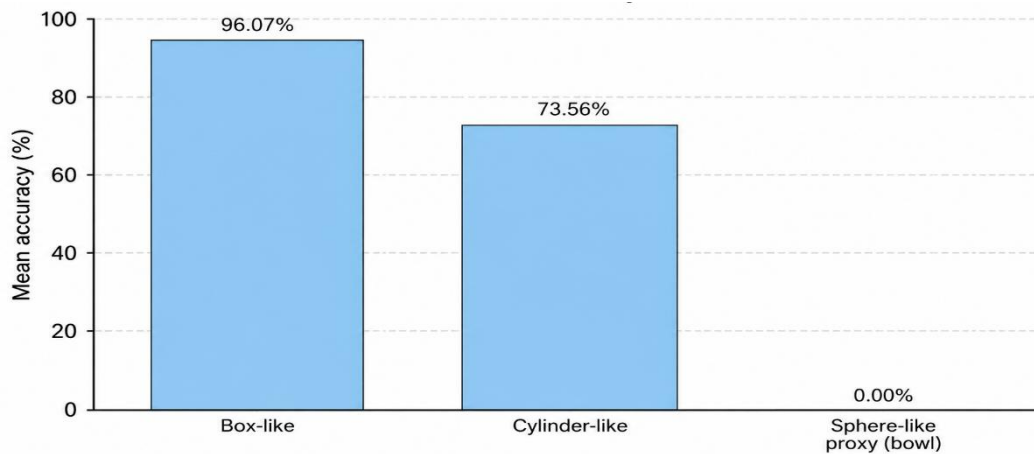


Appendix Figure II. Mean accuracy on the ModelNet40 primitive-like subset. Ten-vote evaluation was applied only to the mixed model after it had been selected as the best single-pass condition.

Appendix Table I2. Final mixed-model ModelNet40 results using ten-vote test-time evaluation.

Metric	Result
Checkpoints evaluated	9
Samples	486
Votes per sample	10
Mean accuracy	81.00%
Median accuracy	82.30%
Standard deviation	4.37%
Best accuracy	84.98%

Ten-vote evaluation increased the mixed-model mean accuracy from 79.81% to 81.00% and the best accuracy from 83.54% to 84.98%. The voting method did not retrain the network. Each object was sampled ten times using different 1000-point subsets, the predicted class probabilities were averaged, and the final class was selected from the averaged probabilities.



Appendix Figure I2. Class-wise ModelNet40 accuracy for the final mixed-trained XYZ-only model. Bowl is retained only as a weak sphere-like proxy.

Appendix Table I3. Class-wise ModelNet40 results for the final mixed model.

Mapped primitive	Mean accuracy (%)	Interpretation
Box-like	96.07	Strongest mapped class
Cylinder-like	73.56	Moderate transfer
Sphere-like proxy (bowl)	0.00	Weak proxy; not a true sphere

ModelNet40 does not contain a true sphere, ball or similar closed spherical category. Bowl was therefore used only as a weak sphere-like proxy. Across 180 bowl predictions, 171 were classified as cylinder and none were classified as sphere. This should mainly be treated as a limitation of the manual class mapping rather than as evidence of poor performance on true spherical objects. YCB-28 provides the more suitable evaluation of the sphere class.

Appendix J. Extended RProp optimiser results

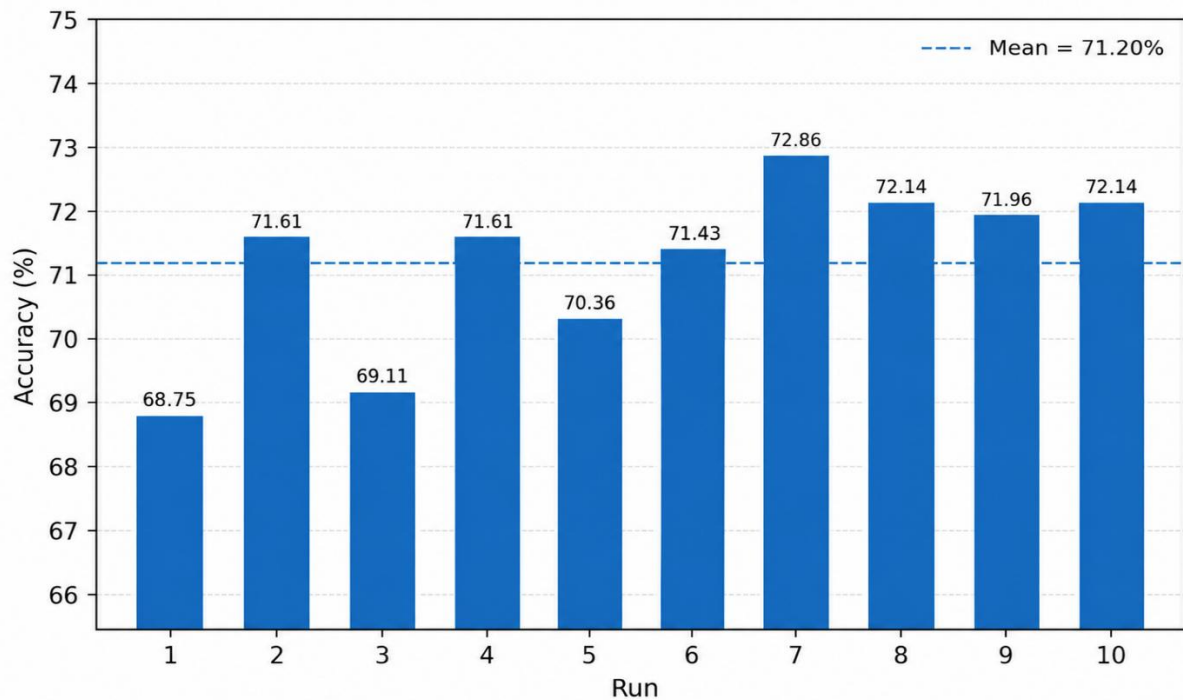
RProp was evaluated as a supplementary optimiser comparison after the main AdamW experiments. The final experiment comprised 10 independent runs, with 71.20% mean YCB-28 accuracy, 71.61% median accuracy and 72.86% best accuracy.

Appendix Table J1. RProp YCB-28 results across ten independent runs.

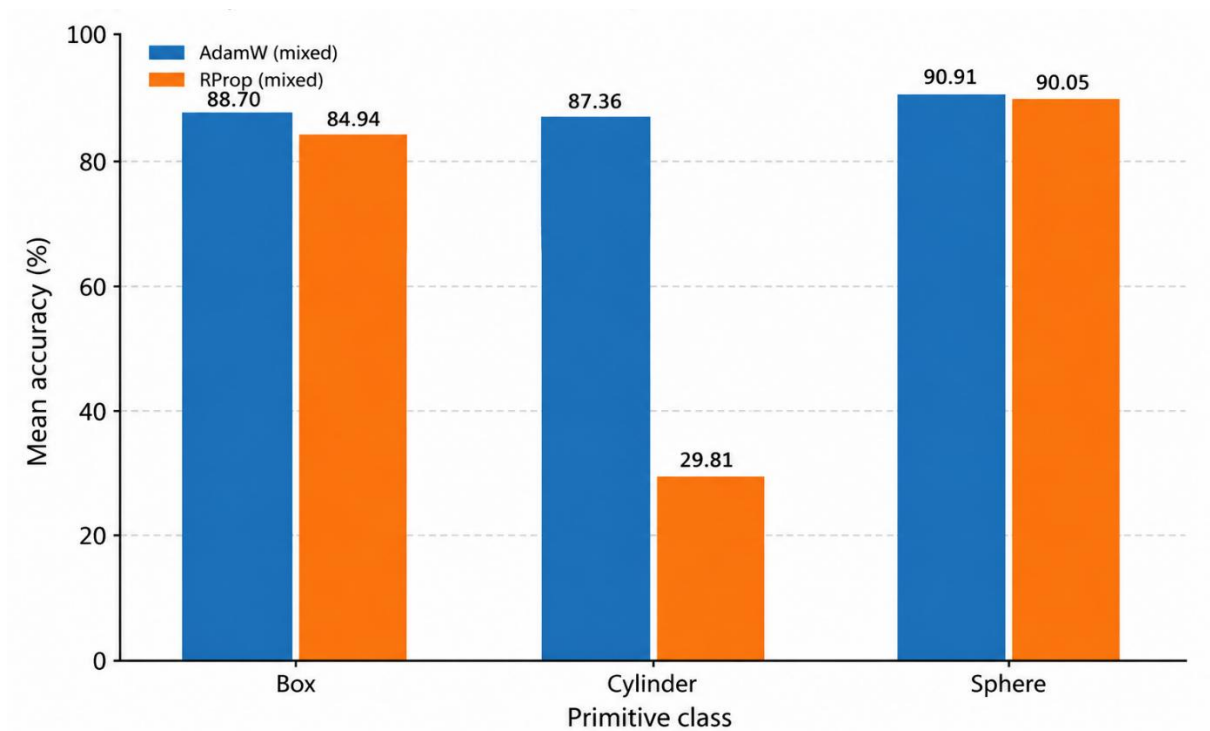
Run	Accuracy (%)	Mean confidence
1	68.75	0.7793
2	71.61	0.7836
3	69.11	0.7799
4	71.61	0.8019
5	70.36	0.8243
6	71.43	0.8010
7	72.86	0.8065
8	72.14	0.7811
9	71.96	0.7732
10	72.14	0.7821

Appendix Table J2. Five-number summary and descriptive statistics for RProp YCB-28 accuracy.

Minimum (%)	Q1 (%)	Median (%)	Q3 (%)	Maximum (%)	Mean (%)	SD (%)
68.75	70.63	71.61	72.10	72.86	71.20	1.35



Appendix Figure J1. RProp YCB-28 accuracy across ten independent runs. The dashed line indicates the mean accuracy.



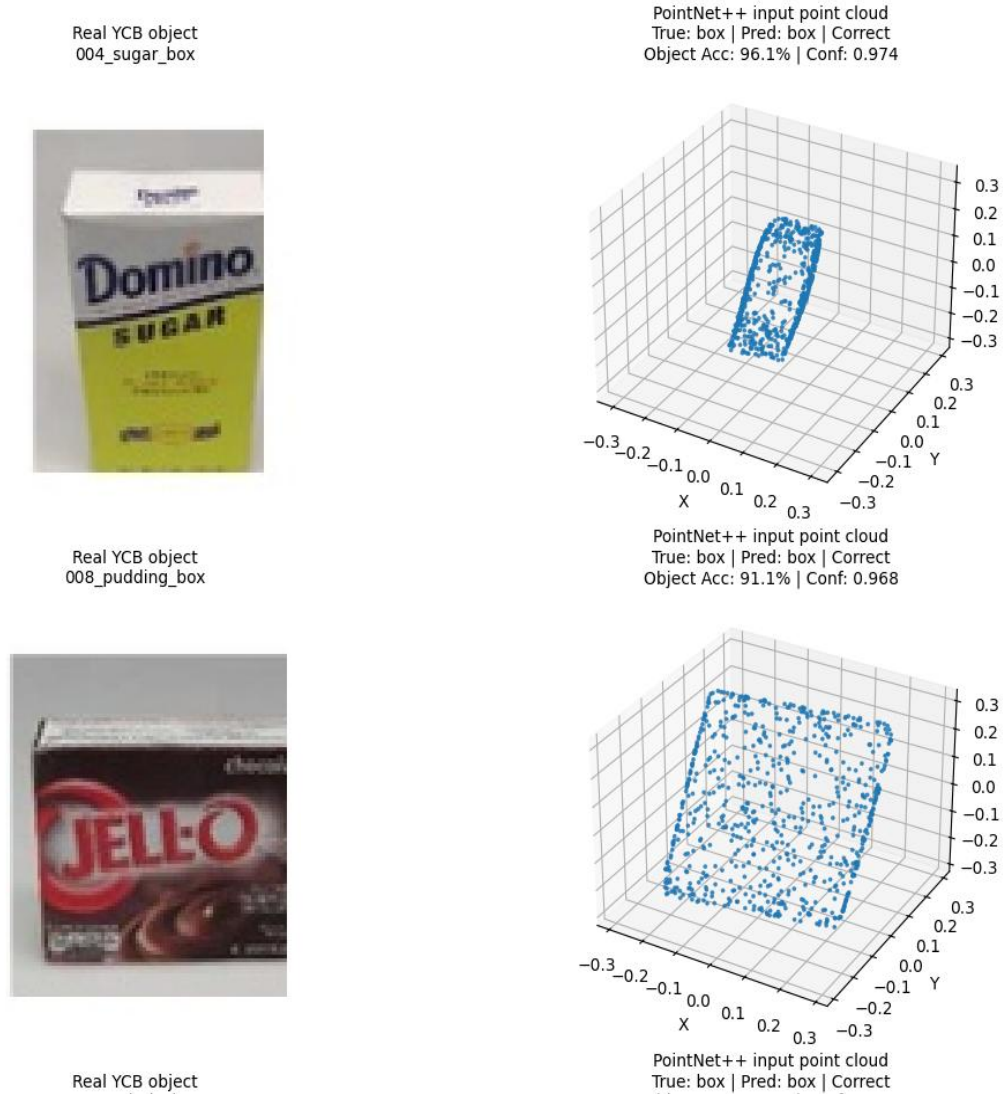
Appendix Figure J2. Class-wise YCB-28 accuracy for the final mixed AdamW and RProp models.

The final class-wise results show that the main weakness of the RProp configuration was cylinder recognition, which achieved 29.81% mean accuracy. Box recognition remained relatively strong at 84.94%, while sphere recognition reached 90.05%, close to the final AdamW sphere result.

Appendix K. Supplementary Real-Object and Point-Cloud Visualisations

This appendix provides qualitative comparisons between the real YCB-28 objects and the corresponding sampled XYZ point clouds presented to the final PointNet++ classifier. In each comparison, the real object is shown alongside its normalised point-cloud representation, with the true primitive class, predicted class, object-level accuracy and confidence reported in the visual annotation. These figures complement the quantitative YCB-28 results by showing how real object geometry is represented after sampling and preprocessing.

The YCB-28 object identities follow the primitive mapping used in this project [9,13]. The point-cloud plots and prediction annotations were generated during the final mixed-trained XYZ-only AdamW evaluation. Representative comparisons are shown below.

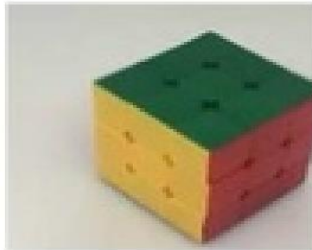


Appendix Figure K1. YCB-28 real-object and PointNet++ input point-cloud comparisons for box-like objects: sugar box and pudding box.

Real YCB object
061_foam_brick



Real YCB object
077_rubiks_cube



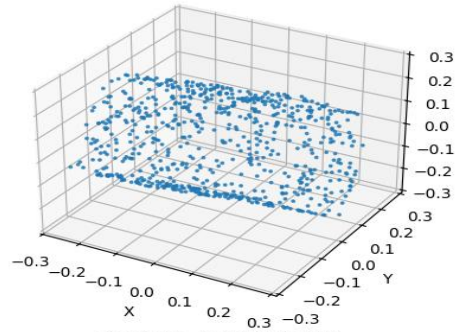
Real YCB object
010_potted_meat_can



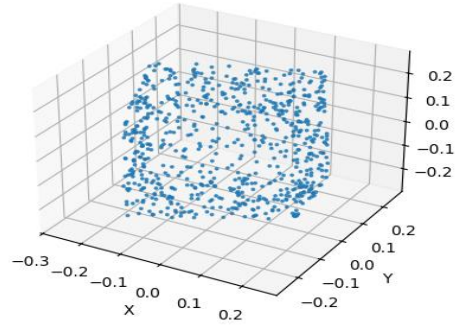
Real YCB object
026_sponge



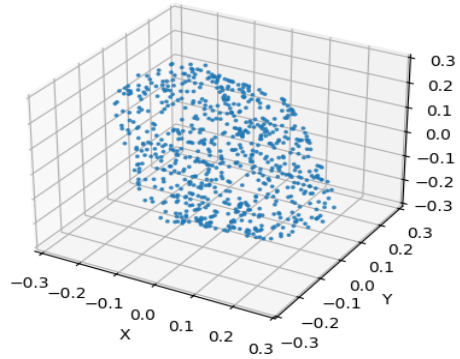
PointNet++ input point cloud
True: box | Pred: box | Correct
Object Acc: 100.0% | Conf: 0.984



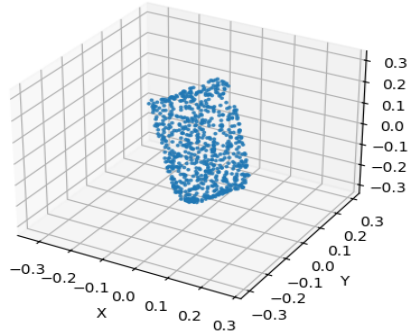
PointNet++ input point cloud
True: box | Pred: box | Correct
Object Acc: 97.8% | Conf: 0.973



PointNet++ input point cloud
True: box | Pred: box | Correct
Object Acc: 93.3% | Conf: 0.966



PointNet++ input point cloud
True: box | Pred: box | Correct
Object Acc: 35.0% | Conf: 0.960

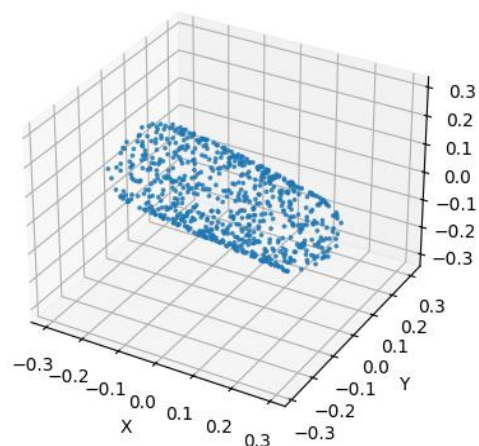


Appendix Figure K2. YCB-28 real-object and PointNet++ input point-cloud comparisons for additional box-like objects: foam brick, Rubik's cube, potted meat can and sponge.

Real YCB object
001_chips_can



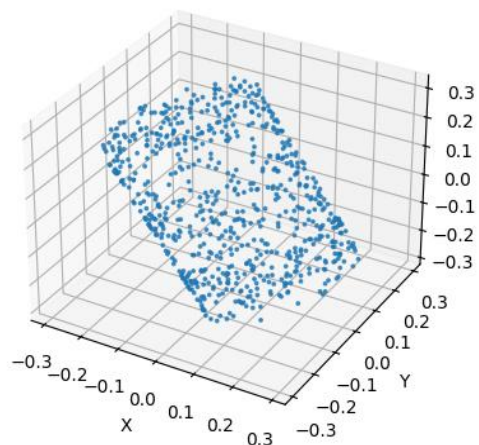
PointNet++ input point cloud
True: cylinder | Pred: cylinder | Correct
Object Acc: 100.0% | Conf: 0.986



Real YCB object
002_master_chef_can



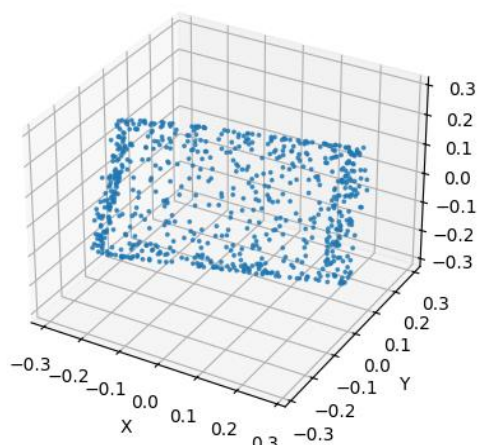
PointNet++ input point cloud
True: cylinder | Pred: cylinder | Correct
Object Acc: 98.9% | Conf: 0.976



Real YCB object
005_tomato_soup_can



PointNet++ input point cloud
True: cylinder | Pred: cylinder | Correct
Object Acc: 91.7% | Conf: 0.971



Appendix Figure K3. YCB-28 real-object and PointNet++ input point-cloud comparisons for cylinder-like objects: chips can, master chef can and tomato soup can.

Real YCB object
019_pitcher_base



Real YCB object
025_mug



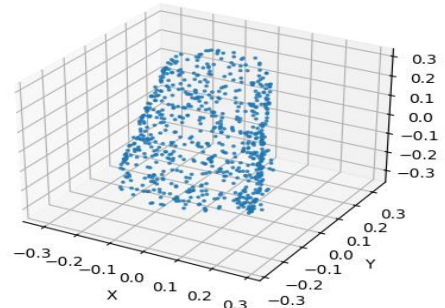
Real YCB object
040_large_marker



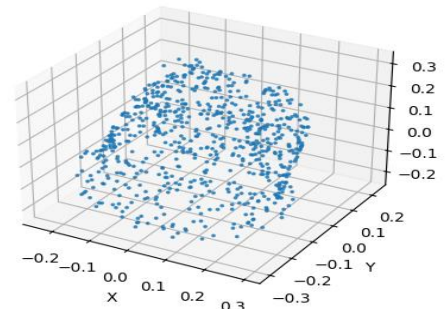
Real YCB object
065_a_cups



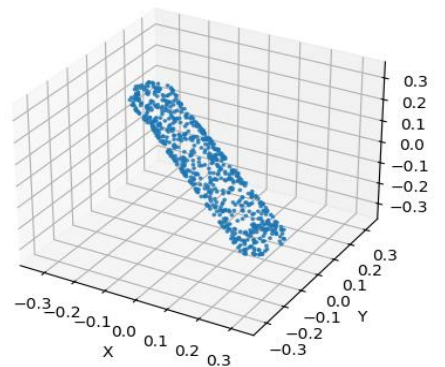
PointNet++ input point cloud
True: cylinder | Pred: cylinder | Correct
Object Acc: 98.9% | Conf: 0.979



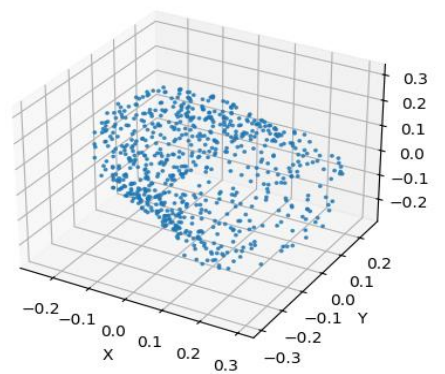
PointNet++ input point cloud
True: cylinder | Pred: cylinder | Correct
Object Acc: 91.7% | Conf: 0.972



PointNet++ input point cloud
True: cylinder | Pred: cylinder | Correct
Object Acc: 32.2% | Conf: 0.841



PointNet++ input point cloud
True: cylinder | Pred: cylinder | Correct
Object Acc: 85.6% | Conf: 0.974

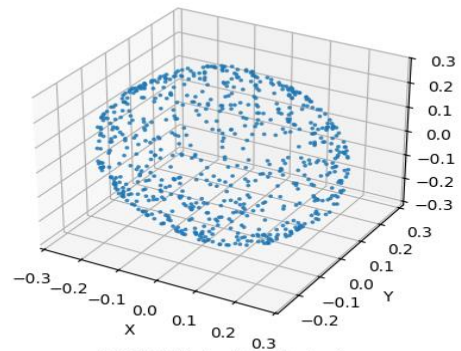


Appendix Figure K4. YCB-28 real-object and PointNet++ input point-cloud comparisons for cylinder-like objects: pitcher base, mug, large marker and cups.

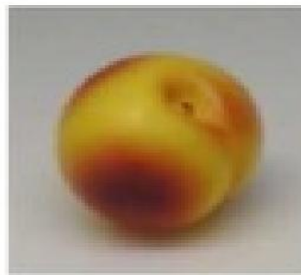
Real YCB object
014_lemon



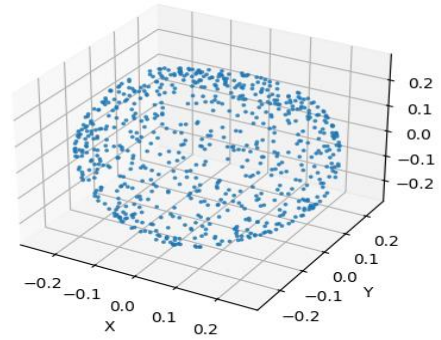
True: sphere | Pred: sphere | Correct
Object Acc: 100.0% | Conf: 0.979



Real YCB object
015_peach



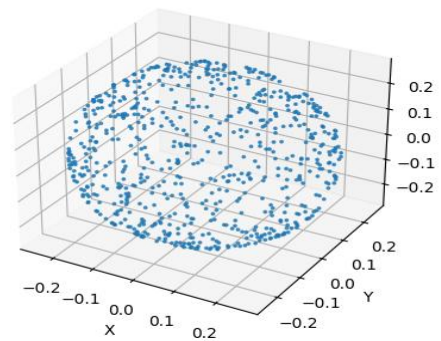
PointNet++ input point cloud
True: sphere | Pred: sphere | Correct
Object Acc: 100.0% | Conf: 0.981



Real YCB object
017_orange



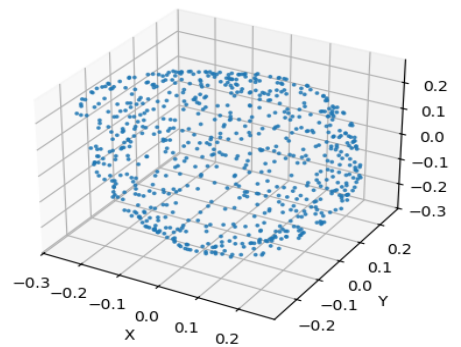
PointNet++ input point cloud
True: sphere | Pred: sphere | Correct
Object Acc: 100.0% | Conf: 0.983



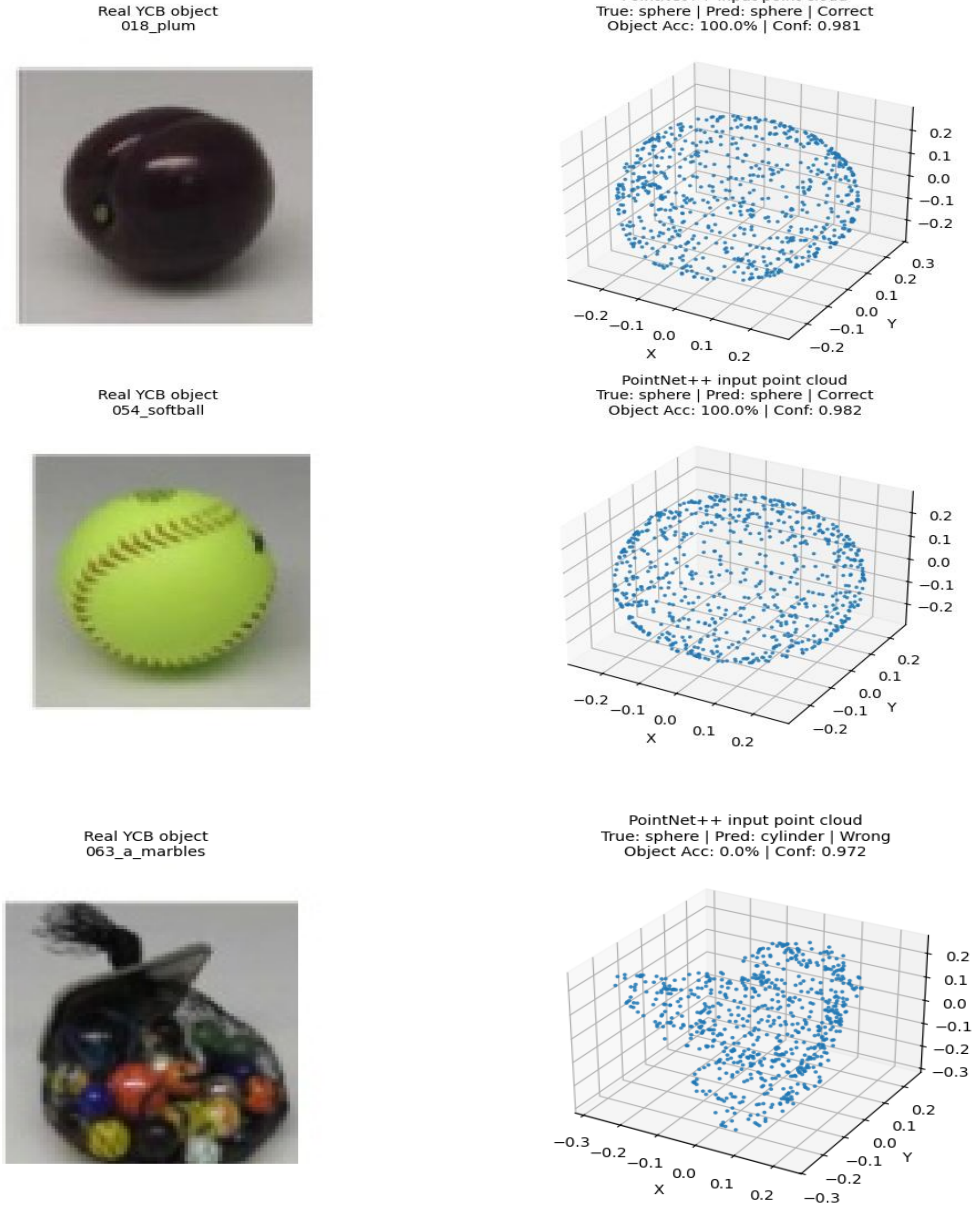
Real YCB object
012_strawberry



PointNet++ input point cloud
True: sphere | Pred: sphere | Correct
Object Acc: 100.0% | Conf: 0.979



Appendix Figure K5. YCB-28 real-object and PointNet++ input point-cloud comparisons for sphere-like objects: lemon, peach, orange and strawberry.



Appendix Figure K6. YCB-28 real-object and PointNet++ input point-cloud comparisons for sphere-like objects: plum, softball and the marbles failure case.

Qualitative interpretation. The paired visualisations show that many YCB objects retain a dominant geometric structure that corresponds reasonably well to the synthetic box, cylinder or sphere classes after point sampling and normalisation. The comparison also highlights the limitations of the three-class abstraction. The large marker is substantially thinner and more elongated than the APS cylinders, the sponge has an irregular rounded cuboid form, and the marbles object is a compound collection rather than a single sphere. These geometric differences are consistent with the lower object-level accuracies reported for these cases in Appendix D.

Source note: the real-object photographs correspond to objects from the YCB object and model set [13] and the YCB-28 mapping used in the reference primitive-recognition study [9]. The sampled point-cloud visualisations, predictions, accuracies and confidence annotations were generated as part of this project's YCB-28 evaluation.